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Institute of electrical and optical communications

Advanced optical fiber communications

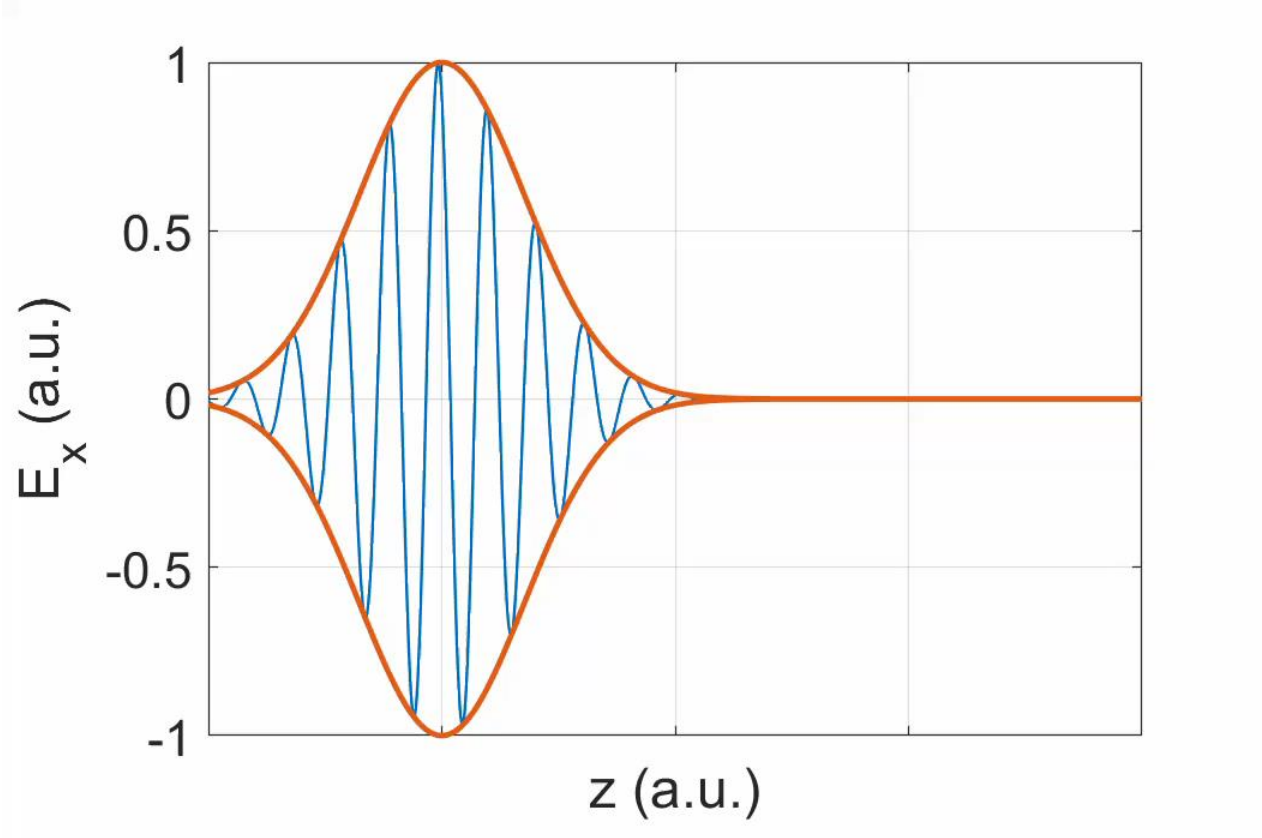
Wintersemester 2025/26

Lecture 3



Prof. Dr.-Ing. Georg Rademacher

Chromatic Dispersion



Chromatic Dispersion

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$$v_{ph} = \frac{k_0 \cdot c}{k_0 \cdot n} = \frac{c}{n}$$

$$v_{gr} = \frac{d\omega}{d\beta} \quad , \quad \beta = k_0 \cdot n \quad , \quad k_0 = \frac{\omega}{c}$$

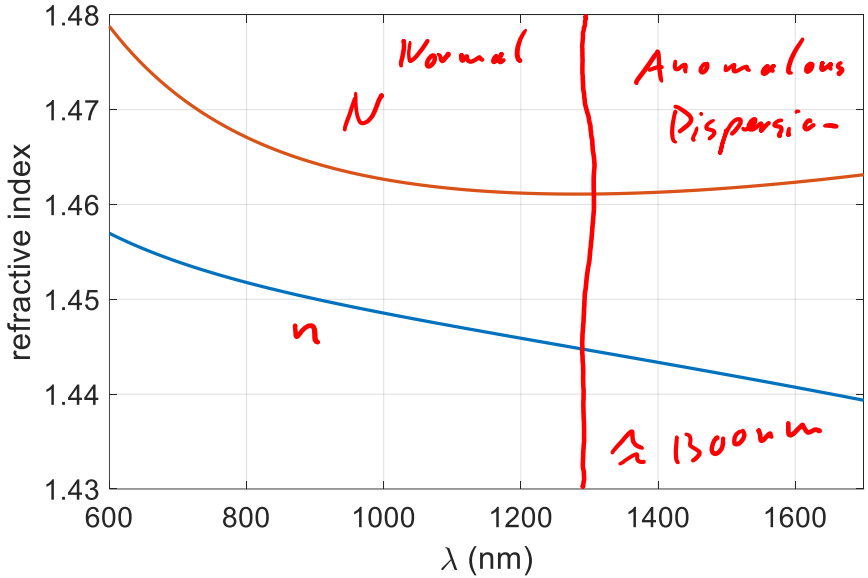
$$v_{gr} = \frac{c}{N}$$

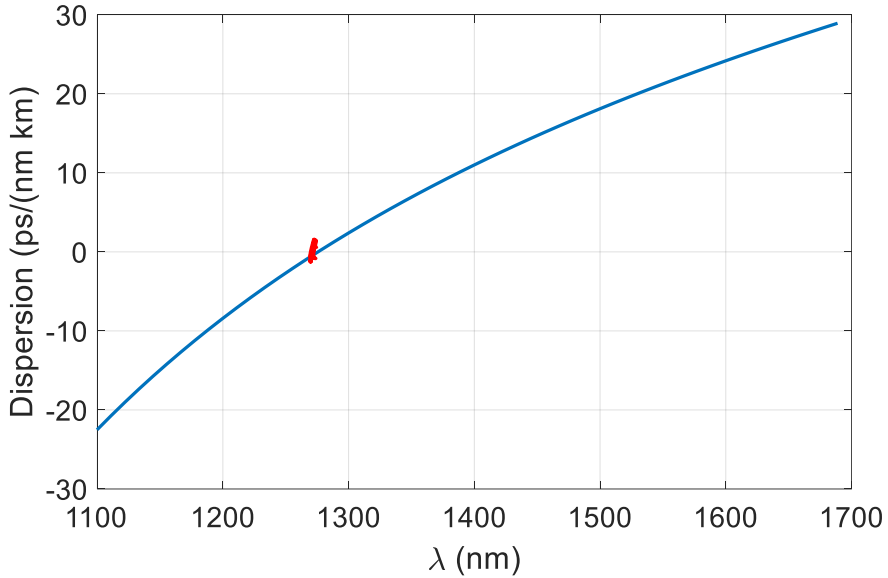
$$D_M = \frac{1}{c} \cdot \frac{dN}{d\lambda} = -\frac{\lambda}{c} \frac{d^2 n(\lambda)}{d\lambda^2}$$

Chromatic Dispersion

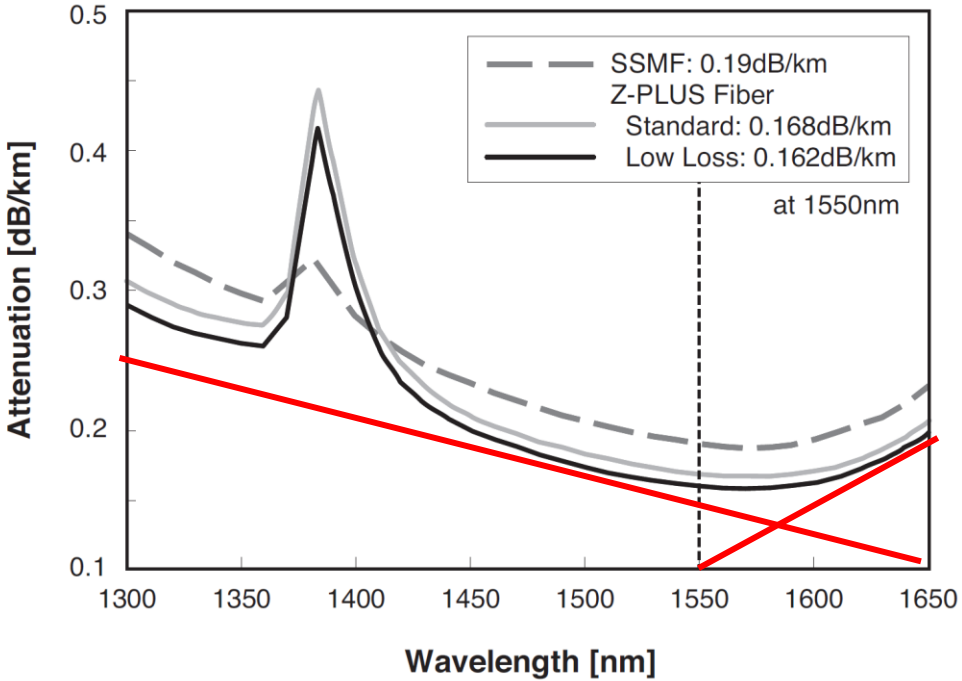
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SiO_2 glass





Attenuation



Yamamoto et al, Low-Loss and Low-Nonlinearity Pure-Silica-Core Fiber for Large Capacity Transmission, SEI Technical review, 2013

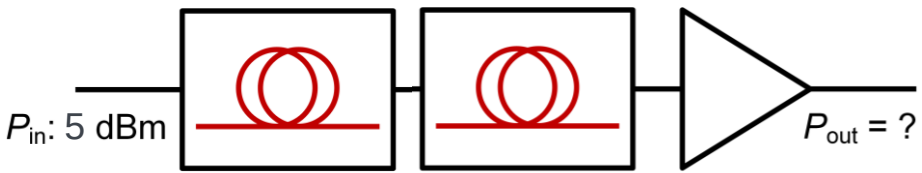
Concept Test

Concept test

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What is the output power of the following optical channel?

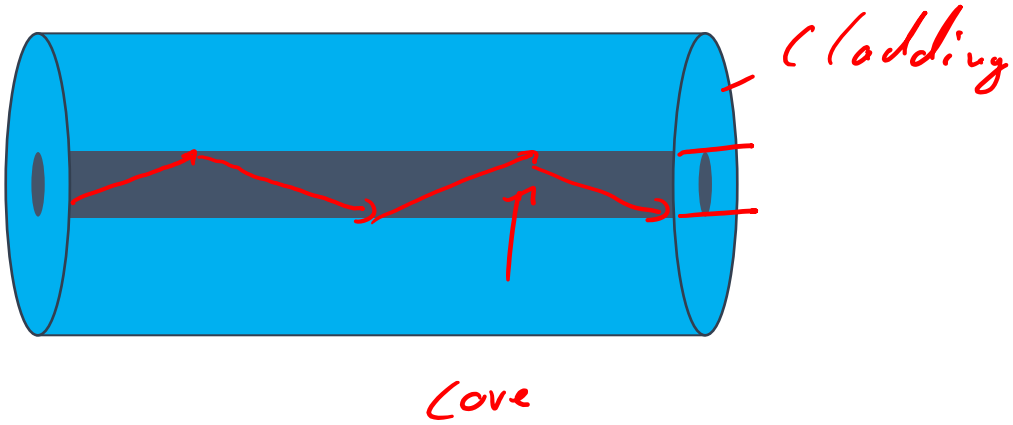
- **A** -1 dBm
- **B** -11 dBm
- **C** -13 dBm



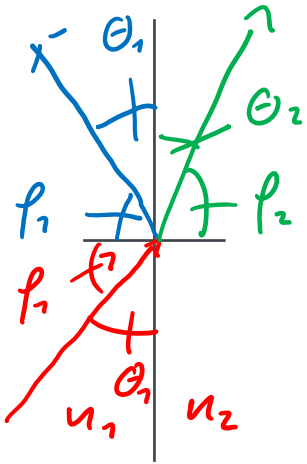
https://ilias3.uni-stuttgart.de/vote/TGI4

Plane waves on dielectric interfaces

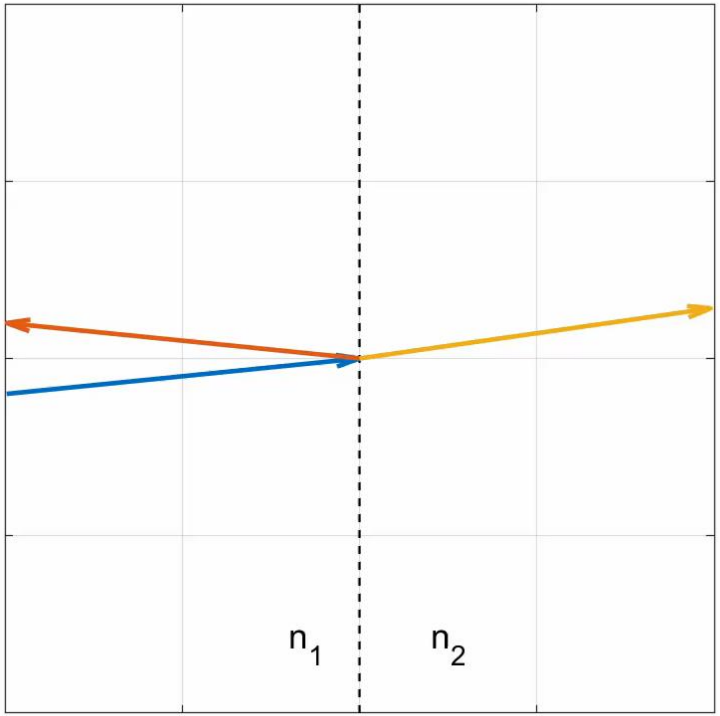
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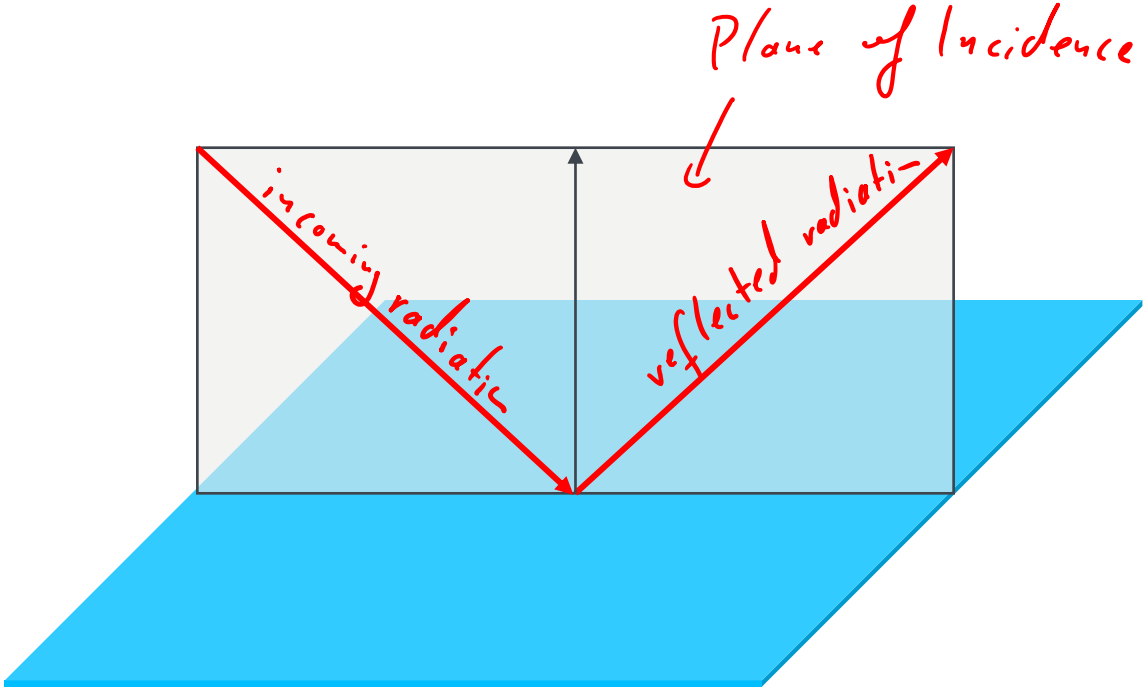
Snell's law



$$\frac{n_1}{n_2} = \frac{\cos(\theta_2)}{\cos(\theta_1)} = \frac{\sin(\rho_2)}{\sin(\rho_1)}$$



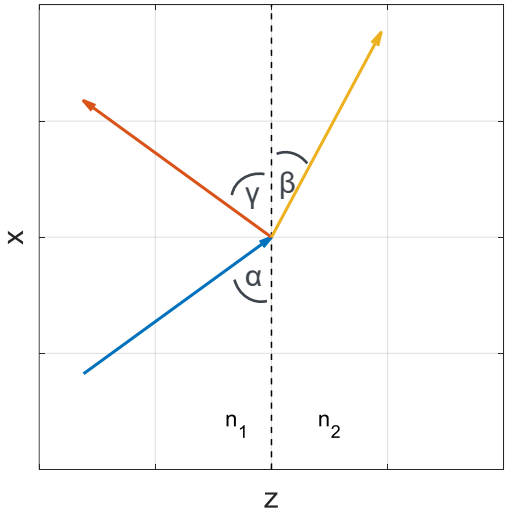
Plane of insidence



Bonus Test

Bonus test

The incident angle is $\alpha = \frac{\pi}{5}$. How large is γ?



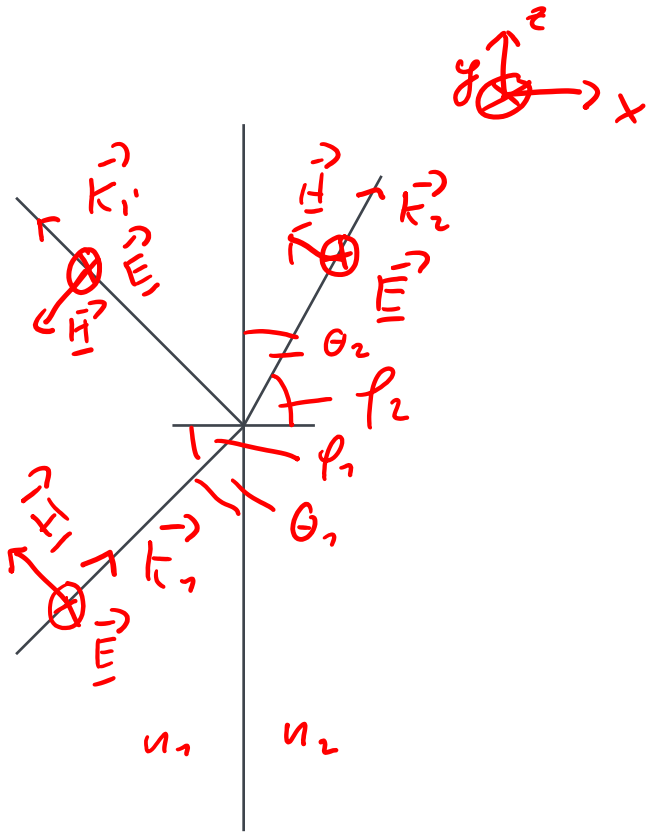
A $\frac{\pi}{5}$

B $\frac{n_1}{n_2}\frac{\pi}{5}$

C $\frac{n_2}{n_1}\frac{\pi}{5}$

https://ilias3.uni-stuttgart.de/goto.php?target=xlvo_1_pin_Q2YS

Plane waves on dielectric interfaces



$\vec{E}$ is orthogonal to plane of incidence.

$$(\vec{k}_1)^2 = (\vec{k}_{1,1})^2 = k_0^2 n_1^2$$

$$(\vec{k}_2) = k_0^2 n_2^2 \qquad \vec{k}_1 = \begin{pmatrix} k_{x1} \\ 0 \\ k_z \end{pmatrix}$$

$$|\vec{k}_1|^2 = k_{x1}^2 + k_z^2 = k_0^2 n_1^2$$

$$\cos(\theta_1) = \frac{k_z}{|\vec{k}_1|} = \frac{k_z}{k_0 n_1}$$

Plane waves on dielectric interfaces

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$$\underline{E}_y = \underline{E}_{y1} \exp(-j k_{x1} \cdot x - j k_z \cdot z)$$

$$\underline{H}_z = \underline{E}_y \frac{\sin(\theta)}{Z_{F1}}$$

$$\underline{H}_x = -\underline{E}_y \frac{\cos(\theta)}{Z_{F1}}$$

$$Z_{F1} = \frac{1}{n_1} \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$Z_{F2} = \frac{1}{n_2} \cdot \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

tangential components must be continuous!

z-components of incident, reflected and refract must be equal

-> z-component of $\vec{k_1}$, $\vec{k_1'}$ & $\vec{k_2}$ = k_z

Plane waves on dielectric interfaces

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$$\frac{k_z}{|\vec{k}_2|} = \cos(\theta_2) = \frac{k_z}{k_0 n_2} \rightarrow k_z = k_0 n_1 \cos(\theta_1) = k_0 n_2 \cos(\theta_2)$$

Snell's law!

$$\cos(\theta_1) > \frac{n_2}{n_1} \quad , \quad \sin(\theta_1) > \frac{n_2}{n_1}$$

$$\sin(\theta_{1g}) = \cos(\theta_{1g}) = \frac{n_2}{n_1}$$

$$\theta_1 < \theta_{1g} \quad , \quad \theta_1 > \theta_{1g} \rightarrow \text{total reflection}$$

Reflectivity of plane waves on dielectric interfaces

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$$\underline{r}_E = \frac{\underline{E}_{y1'}}{\underline{E}_{y1}}$$

E-field is orthogonal to plane of incidence

$$\underline{E}_{y1} + \underline{E}_{y1'} = \underline{E}_{y2} \quad , \quad \underline{H}_{z1} + \underline{H}_{z1'} = \underline{H}_{z2}$$

$$\underline{Z}_{1E} = \frac{\underline{E}_{y1}}{\underline{H}_{z1}} = -\frac{\underline{E}_{y1'}}{\underline{H}_{z1}} = \frac{1}{n_1 \sin(\theta_1)} \sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{k_0}{k_{x1}} \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$\underline{Z}_{2E} = \frac{\underline{E}_{y2}}{\underline{H}_{z2}} = \frac{1}{n_2 \sin(\theta_2)} \sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{k_0}{k_{x2}} \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

Plane waves on dielectric interfaces

$$\underline{E}_{y1}\frac{\underline{Z}_{2E}}{\underline{Z}_{1E}} - \underline{E}_{y1}\frac{\underline{Z}_{2E}}{\underline{Z}_{1E}} = \underline{E}_{y2}$$

$$\underline{r}_E = \frac{\underline{Z}_{2E} - \underline{Z}_{1E}}{\underline{Z}_{2E} + \underline{Z}_{1E}} = \frac{n_1\cos(\varphi_1) - n_2\cos(\varphi_2)}{n_1\cos(\varphi_1) + n_2\cos(\varphi_2)}$$

$$\rightarrow \underline{r}_E = \frac{\sin(\varphi_2 - \varphi_1)}{\sin(\varphi_2 + \varphi_1)}$$

Plane waves on dielectric interfaces

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$$\underline{H}_y = \underline{H}_{y1} \exp(-j k_{x1} x - j k_z z)$$

$$\underline{E}_z = -\underline{H}_y \cdot \sin(\theta_1) \cdot Z_{F1}$$

$$\underline{E}_x = \underline{H}_y \cdot \cos(\theta_1) \cdot Z_{F1}$$

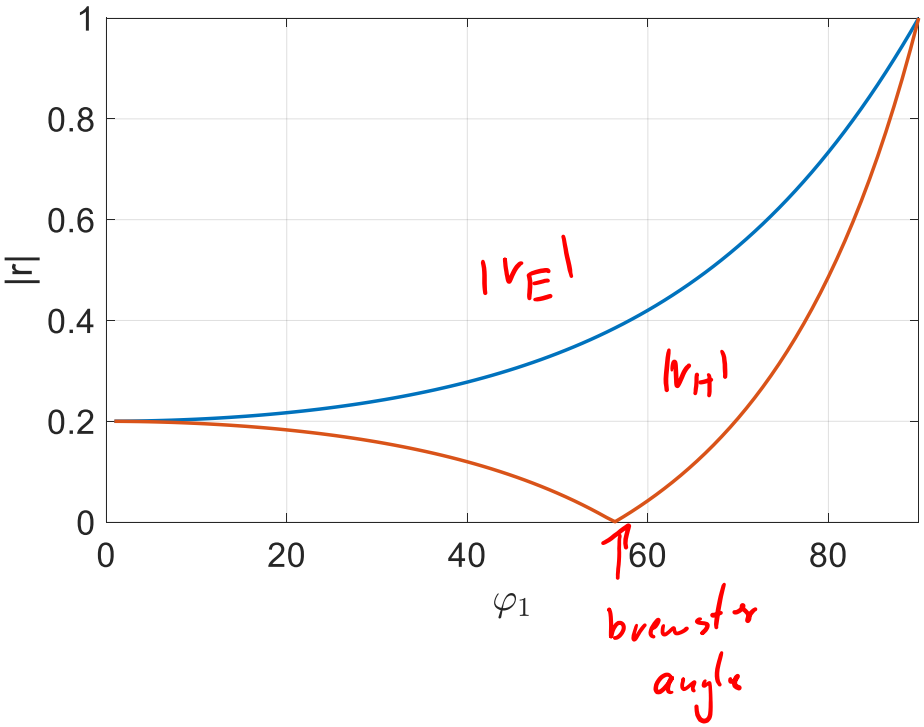
$$\underline{r}_H = \frac{\underline{Z}_{2H} - \underline{Z}_{1H}}{\underline{Z}_{2H} + \underline{Z}_{1H}} = \frac{n_1 \sin(\theta_2) - n_2 \sin(\theta_1)}{n_1 \sin(\theta_2) + n_2 \sin(\theta_1)}$$

$$\underline{r}_H = \frac{\cos(\beta_2 + \beta_1) \cdot \sin(\beta_2 - \beta_1)}{\cos(\beta_2 - \beta_1) \cdot \sin(\beta_2 + \beta_1)}$$

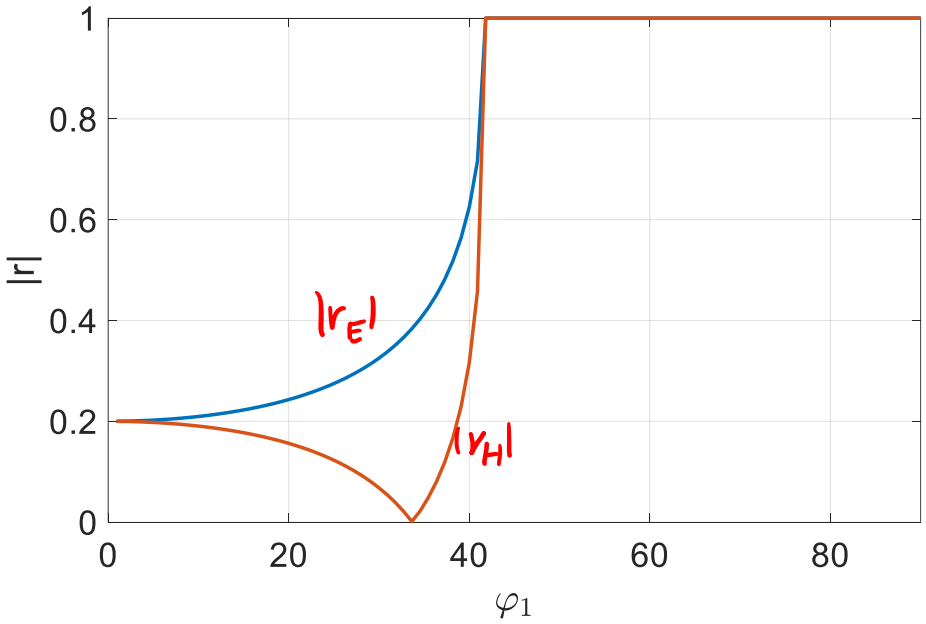
Plane waves on dielectric interfaces

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$n_1 < n_2$



$n_1 > n_2$



Plane waves on dielectric interfaces

$\varphi_1 = \varphi_{1B} \rightarrow r_{\perp} = 0$

$\varphi_{1B} \rightarrow \varphi_1 + \varphi_2 = \pi/2$

$\rightarrow \tan(\varphi_{1B}) = \frac{n_2}{n_1}$

Concept Test

Concept test

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Is it possible to reach the maximum transmission of 1 for $n_1 \neq n_2$? If yes, which field vector has to be parallel to the plane of reflection?

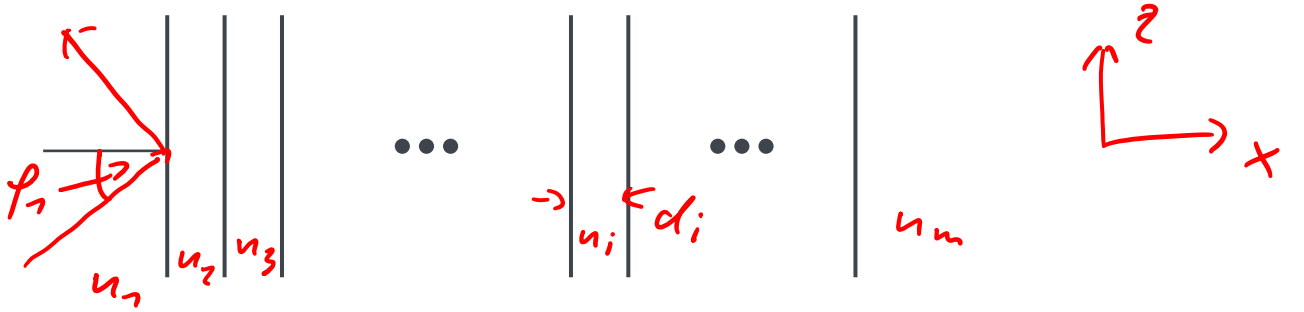
A Yes, electric field.

B Yes, magnetic field.

C No, it‘s not possible.

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Reflection on multiple layers



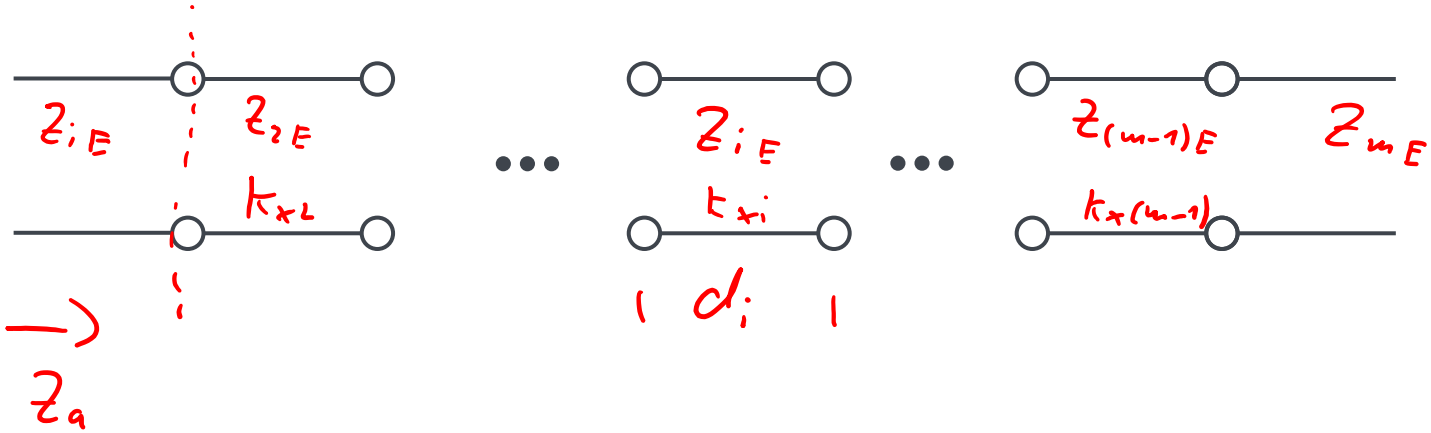
$$k_z = k_0 n_1 \sin(\rho_1)$$

$$k_{xi} = k_0 \sqrt{n_i^2 - n_1^2 \sin(\rho_1)}$$

$$Z_{iE} = \frac{k_0}{k_{xi}} \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

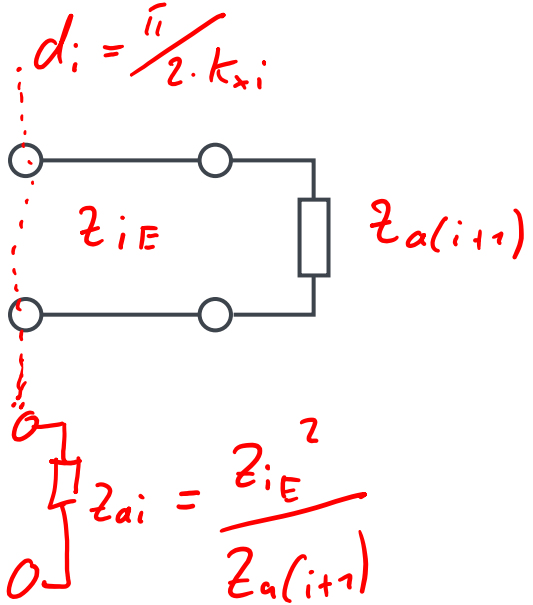
$$Z_{iH} = \frac{k_{xi}}{n_i^2 k_0} \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

Reflection on multiple layers



Reflection on multiple layers

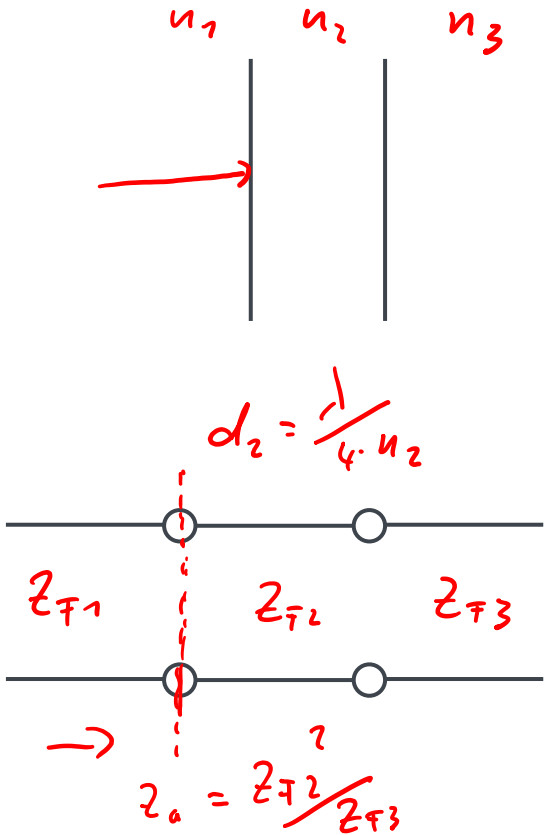
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$$\frac{\lambda}{4} : k_{xi} \cdot d_i = \frac{\lambda}{2}$$

$$r_E = \frac{z_a - z_{1E}}{z_a + z_{1E}}$$

Example: $\lambda/4$ layer for anti-reflection coatings



$$(\rho_1) = 0$$

$$Z_{iE} = Z_{iH} = Z_F = \frac{1}{n_i} \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

Example: $\lambda/4$ layer for anti-reflection coatings

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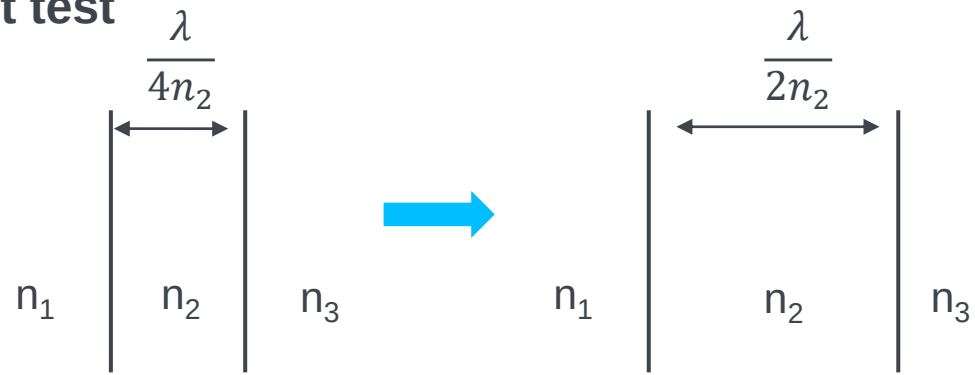
$$Z_a = \frac{Z_{F2}^2}{Z_{F3}} = \frac{n_3}{n_2^2}\sqrt{\frac{\mu_0}{\varepsilon_0}} \quad , \quad \underline{r}_E = \underline{r}_H = 0$$

$$Z_{F2} = \sqrt{Z_{F1} \cdot Z_{F3}} \Rightarrow n_2 = \sqrt{n_1 \cdot n_3} \Rightarrow Z_a = Z_{F1}$$

$$d_2 \cdot k_{x2} = d_2 \cdot k_0 \cdot n_2 = d_2 \cdot n_2 \frac{2\pi}{\lambda} \stackrel{!}{=} \frac{\pi}{2} \quad \Rightarrow d_2 = \frac{\lambda}{4 \cdot n_2}$$

Concept Test

Concept test



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How is the functionality of the anti-reflection coating affected, if the length of the intermediate layer is $\lambda/2$ instead of $\lambda/4$?

A It‘s not affected.

B All the incident light will be reflected.

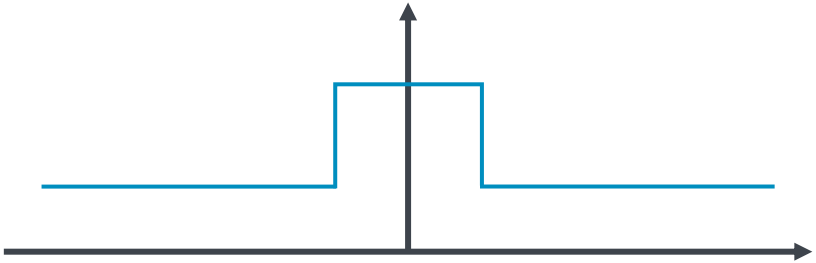
C The intermediate layer becomes transparent.

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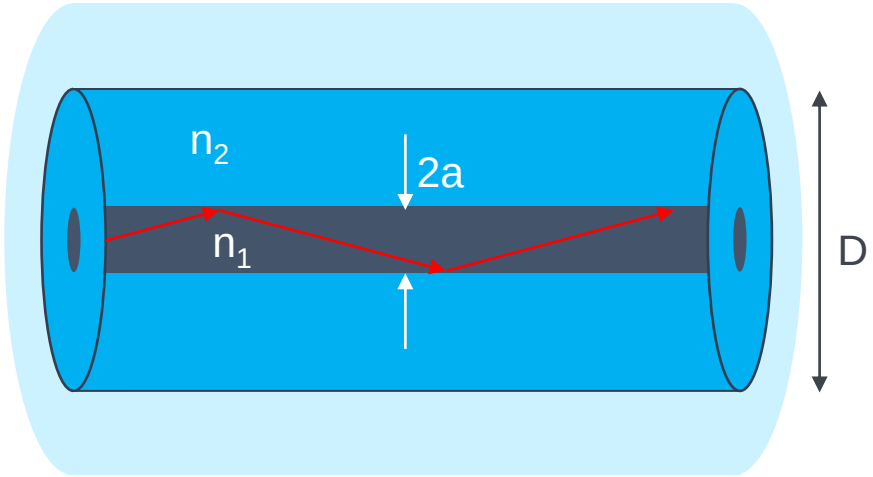
Step-index fiber



Step-index fiber



Step-index fiber



Electromagnetc fields in step-index fiber

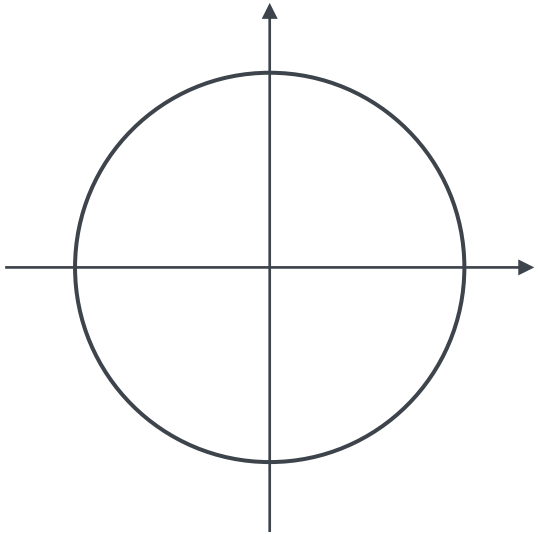
Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

Electromagnetc fields in step-index fiber

Boundary conditions in step-index fiber



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Thank you!

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